79. Rotary Heart Assist Devices

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The left ventricular assist device (LVAD) is a mechanical device implanted in patients with congestive heart failure to assist the heart in pumping blood through the circulatory system. The latest generation of this device is comprised of a rotary pump which is generally much smaller, lighter, and quieter than the first-generation conventional pulsatile-type pump. The rotary pump is controlled by varying the rotor (or impeller) speed to adjust the amount of blood flow through the LVAD. If the patient is in a health care facility, the pump speed can be adjusted manually by a trained clinician to meet the patient's blood needs. However, an important challenge facing the increased use of these devices is the desire to allow the patient to return home. The development of an appropriate feedback controller that is capable of automatically adjusting the pump speed is therefore a crucial step in meeting this challenge. In addition to being able to adapt to changes in the patient's daily activities by automatically regulating the pump speed, the controller must also be able to prevent the occurrence of excessive pumping. This dangerous phenomenon, known as suction, may cause collapse of the ventricle and damage to the heart muscle. In order to be able to develop such a controller based on modern control theory an appropriate mathematical model of a combined cardiovascular system and LVAD must first be developed. In this chapter, we develop such a model. The model is dynamic, time-varying, and consists of six coupled nonlinear differential equations. The time variation occurs over four consecutive intervals representing the contraction, ejection, relaxation, and filling phases of the left ventricle. The LVAD in the model along with its inlet and outlet cannulae are represented by a nonlinear differential equation which relates the pump rotational speed and pump flow to the pressure difference across the pump. Suction is accounted for by adding a nonlinear resistance in the LVAD model when the pressure in the left ventricle drops below a specified threshold. Using this model we discuss some of the challenges faced in the development of: (1) an appropriate feedback controller for the LVAD, and (2) an effective algorithm for detection of suction in the left ventricle.

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Heart transplantation has now been recognized as the best therapy for patients with end-stage congestive heart failure. However, potential recipients often wait long periods of time (300 days or more on the average) before a suitable donor heart becomes available, and many of these candidates, 20–30%, will die while awaiting heart transplantation. Consequently, the medical community has placed increased emphasis on the use of mechanical circulatory assist devices that can substitute for, or enhance, the function of the natural heart while the patient is waiting for the heart transplant [79.1, 2]. A left ventricular assist device (LVAD) is such a device. There are two different types of LVADs. The first is a pulsatile pump producing pulsatile pressures and

flows similar to those produced by the natural heart. The second, and more recent, is a rotary pump that operates continuously attempting to draw blood out of the left ventricle and into the circulation. Rotary pumps are typically, quieter, smaller, and more efficient than pulsatile pumps, and consequently have received considerable acceptance in recent years. Generally speaking, the goal of the LVAD is to assist the native heart in pumping blood so as to provide the patient with as close to a normal lifestyle as possible until a donor heart becomes available or, in some cases, until the patient's heart recovers. In many situations, this means allowing the patient to return home and/or to the workforce.

An important engineering challenge facing the increased use of the rotary LVAD is the development of an appropriate controller for the speed of the rotor (or pump impeller). Such a controller, in addition to being robust and reliable, must be able to adapt to the daily activities and physiological changes of the patient by regulating the pump speed in order to meet the body's requirements for cardiac output (CO) and mean arterial pressure (MAP) [79.3, 4]. Since the rotary pump does not use valves, if its speed is too low, blood may regurgitate back from the aorta to the left ventricle through the pump, resulting in what is known as *backflow*. If the pump speed is too high, the pump will attempt to draw more blood from the ventricle than available, which may cause collapse of the ventricle resulting in a dangerous phenomenon called *suction*. The occurrence of suction must be detected quickly and the pump speed reduced before the heart muscle is damaged. While avoiding these two extremes, the pump speed must also be adjusted continuously, up and down to meet the patient's varying levels of physical activity and emotional changes [79.5–10]. The eventual goal of a pump speed controller is therefore to meet all these requirements so that an LVAD recipient patient could potentially leave the hospital and return home to a normal lifestyle. Given that the pump is continuously interacting with the cardiovascular system, the development of a speed controller that meets the above objectives must therefore be done using tools developed in modern control theory. This cannot be done without first having an appropriate mathematical model for this complex system. The model must be simple enough to be tractable and yet it must be comprehensive enough to capture the essential relationships between the hemodynamic variables and provide the important input and output boundary conditions without the ambiguity of unnecessary state variables. In this chapter, we discuss the development of such a model. We first present and validate an autonomous fifth-order model of the cardiovascular system which emphasizes the pressure–volume relationship of the left ventricle. We then present a first-order model of a rotary LVAD along with its inlet and outlet cannulae and with the pump rotational speed as its control variable. We account for the phenomenon of suction by adding a nonlinear resistance to the inlet cannula model which becomes active when the left ventricular pressure drops below a certain threshold. Finally, we combine these two models into a sixth-order time-varying nonlinear model with the rotational speed of the pump being the only control variable. The time variations in this model are due to the cyclical nature of the ventricle elastance and its changes as a function of time within one cardiac cycle. The nonlinearities are due to the mitral and aortic valves in the cardiovascular model and the suction resistance in the LVAD model. The binary state of each of these valves yield four consecutive phases within one cardiac cycle during which the ventricle contracts, ejects, relaxes, and then fills depending on which valve is open and which is closed. As a result, the mathematical description of the entire model changes as the ventricle crosses from one phase to the next. We will conclude this chapter by outlining some of the challenges in the development of both a feedback speed controller and a suction detection algorithm that need to be overcome before the ultimate goal of the LVAD becomes a reality and LVAD patients are able to return to a normal lifestyle.

79.1 The Cardiovascular Model

A cross-section schematic of the heart illustrating its various components is shown in Fig. 79.1. Each side of the heart has its own atrium and ventricle, and each side controls a different blood circulation path. The left side of the heart controls the systemic circulation of blood throughout the body while the right side controls the pulmonary circulation of blood to the lungs [79.11]. After the blood picks up oxygen in the lungs it returns to the heart entering through the left atrium. The left ventricle pumps the oxygen-rich blood through the aorta to the rest of the body. The blood returns to the heart depleted of oxygen, this time through the right atrium. It

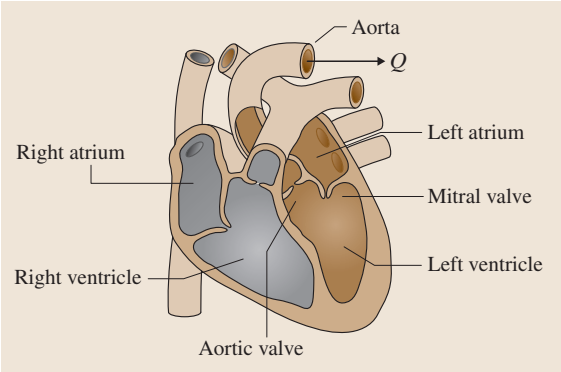


Fig. 79.1 Cross-section of a beating heart

is then pumped to the lungs by the right ventricle. The blood receives oxygen in the lungs, and returns to the left atrium of the heart. Upon entering the left atrium, the two circulatory paths are then repeated. The blood path through the heart and both circulation systems can be seen in Fig. 79.2 [79.12]. Due to the different nature of the two circulation paths, each side of the heart has a different workload. The systemic circulation is much longer than the pulmonary one, and encounters considerable more resistance [79.11]. To handle the longer path, the left ventricle is more elastic than the right, giving it the strength to pump blood at a higher pressure.

The heart is a very complex system that is very difficult to model mathematically. Numerous dynamical state-space models of varying degrees of complexity have been developed in the past fifteen years or so. For example, an eighth-order hybrid model of the heart which subdivides the human circulatory system into a number of lumped parameter blocks that include the pulmonary arterial and venous systems as well as both ventricles has been developed in [79.13]. An 11th-order model which includes the resistance of peripheral blood vessels has been developed in [79.14]. Although a complete heart model can be supplemented with a model of an LVAD, in this chapter we assume that the right ventricle and pulmonary circulation are healthy and normal and as a result their effect on the LVAD, which is connected from the left ventricle to the ascending aorta, can be neglected [79.1, 6]. A fifth-order lumped parameter electric circuit model which can reproduce the left ventricle hemodynamics of the heart [79.15, 16] is shown in Fig. 79.3. In this model, the behavior of the left ventricle is modeled by means of a time-varying capacitance (or compliance) $C(t)$, which is the reciprocal of the ventricle's elastance function $E(t)$. The elastance represents the contractual state of the left ventricle. It relates to the ventricle's pressure and volume [79.17] according to an expression of the form

$$E(t) = \frac{\mathrm{LVP}(t)}{\mathrm{LVV}(t) - V_0} \,, \tag{79.1}$$

where $\mathrm{LVP}(t)$ is the left ventricular pressure, $\mathrm{LVV}(t)$ is the left ventricular volume, and V_0 is a reference volume, which corresponds to the theoretical volume in the ventricle at zero pressure. Several mathematical expressions have been derived to approximate the elastance function $E_\mathrm{H}(t)$ of a healthy heart. In our work, we use the expression

$$E_\mathrm{H}(t) = (E_\mathrm{max} - E_\mathrm{min}) E_\mathrm{n}(t_\mathrm{n}) + E_\mathrm{min} \,, \tag{79.2}$$

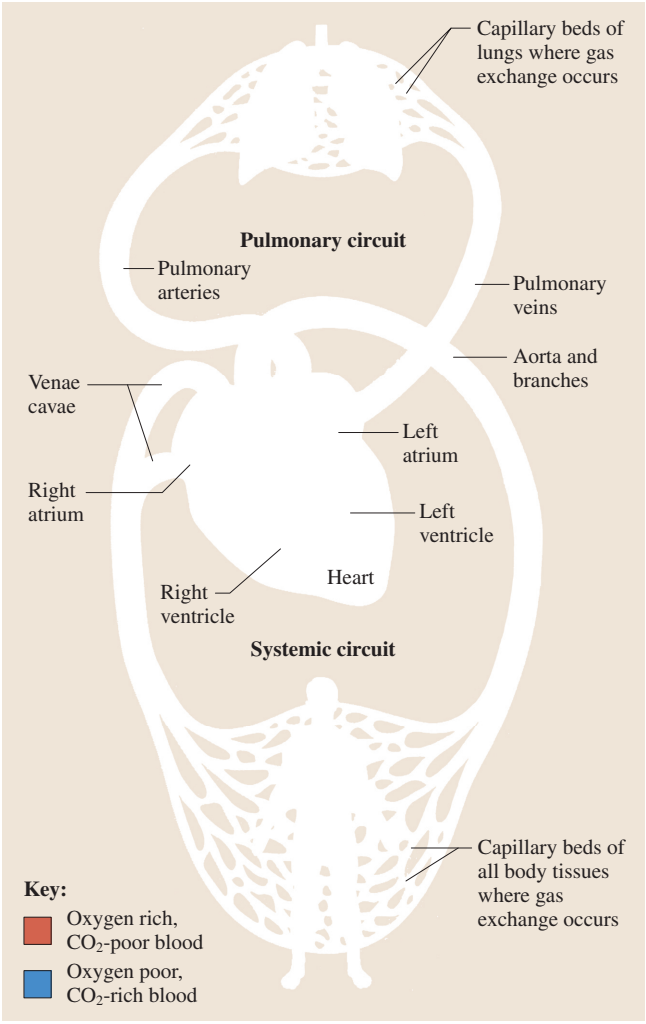


Fig. 79.2 Diagram of systemic and pulmonary circulatory systems (after [79.12])

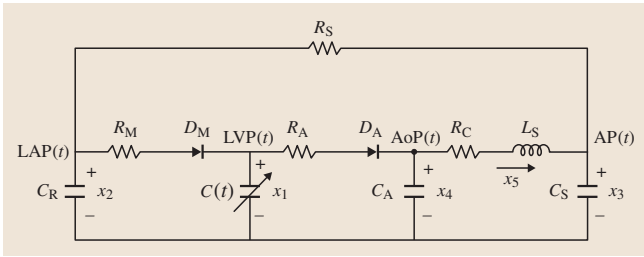


Fig. 79.3 Cardiovascular circuit model

where $E_n(t_n)$ is the so-called *double hill* function represented by the expression [79.18]

$$E_n(t_n) = 1.55 \left(\frac{\left(\frac{t_n}{0.7}\right)^{1.9}}{1+\left(\frac{t_n}{0.7}\right)^{1.9}} \right) \left(\frac{1}{1+\left(\frac{t_n}{1.17}\right)^{21.9}} \right) . \tag{79.3}$$

In the above expression, $E_n(t_n)$ is the normalized elastance, $t_n = t/T_{max}$, $T_{max} = 0.2 + 0.15t_c$, and t_c is the cardiac cycle interval, i. e., $t_c = 60/\text{HR}$, where HR is the heart rate. Notice that $E_H(t)$ is a rescaled version of $E_n(t_n)$ and the constants E_{max} and E_{min} are related to the end-systolic pressure–volume relationship (ESPVR) and the end-diastolic pressure–volume relationship (EDPVR), respectively. Figure 79.4 shows a plot of $E_H(t)$ for a healthy heart with $E_{max} = 2\,\text{mmHg/ml}$, $E_{min} = 0.06\,\text{mmHg/ml}$, and a heart rate of 60 bpm (beats per minute). For a heart with cardiovascular disease, the elastance expression used in our model is modified according to $E(t) = \delta E_H(t)$. That is, the elastance $E_H(t)$ is scaled with a factor $0 < \delta \leq 1$, where $\delta = 1$ represents a healthy heart and smaller values of δ are used to represent cardiovascular disease. The more severe the disease, the smaller the value of δ. Also in the model, preload and pulmonary circulations are represented by the capacitance C_R; the aortic compliance is represented by the capacitance C_A, and afterload is represented by the four-element Windkessel model [79.19] comprising R_C, L_S, C_S, and R_S. Preload, or venous blood returning to the heart, is closely related to the cardiac output (CO) produced by the heart for a given level of contractility. Afterload can be described as the pressure that the left ventricle has to generate in order to eject blood. It refers to the vascular resistance that the heart sees as it pumps blood. The mitral and aortic valves (Fig. 79.1) are represented by two nonideal diodes consisting of a resistance R_M and ideal diode D_M for the mitral valve, and resistance R_A and ideal diode D_A for the aortic valve. In this representation, we have kept the number of model parameters to a minimum while maintaining enough complexity in the model so that it can reproduce the hemodynamics of the left ventricle. Table 79.1 lists the various system parameters and their typical associated values [79.20–22].

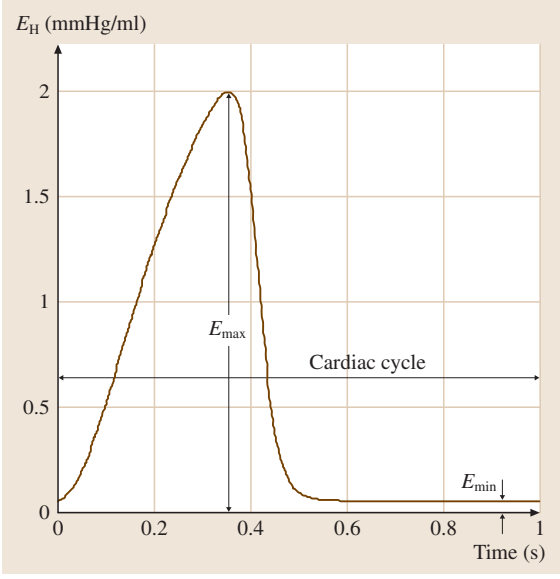


Fig. 79.4 Elastance function $E_H(t) = \frac{1}{C(t)}$ of a healthy heart (cardiac cycle = 60/HR)

Obviously the circuit model in Fig. 79.3 is time-varying because of the capacitance $C(t)$ and nonlinear because of the two diodes. Since each ideal diode has two states: open-circuit (O/C) and short-circuit (S/C),

Table 79.1 Model parameters

Parameters	Value	Physiological meaning
Resistances (mmHg s/ml)		
R_S	1.0000	Systemic vascular resistance (SVR)
R_M	0.0050	Mitral valve resistance
R_A	0.0010	Aortic valve resistance
R_C	0.0398	Characteristic resistance
Compliances (ml/mmHg)		
$C(t)$	Time-varying	Left ventricular compliance
C_R	4.4000	Left atrial compliance
C_S	1.3300	Systemic compliance
C_A	0.0800	Aortic compliance
Inertances (mmHg s^2/ml)		
L_S	0.0005	Inertance of blood in aorta
Valves		
D_M		Mitral valve
D_A		Aortic valve

Table 79.2 Phases of the cardiac cycle

Modes	Valves Mitral	Aortic	Phases
1	Closed	Closed	Isovolumic relaxation
2	Open	Closed	Filling
1	Closed	Closed	Isovolumic contraction
3	Closed	Open	Ejection
–	Open	Open	Not feasible

Table 79.3 State variables in the cardiovascular model

Variables	Name	Physiological meaning (unit)
$x_1(t)$	LVP(t)	Left ventricular pressure (mmHg)
$x_2(t)$	LAP(t)	Left atrial pressure (mmHg)
$x_3(t)$	AP(t)	Arterial pressure (mmHg)
$x_4(t)$	AoP(t)	Aortic pressure (mmHg)
$x_5(t)$	$Q_T(t)$	Total flow (ml/s)

four different circuits representing different phases of the ventricular function can therefore be modeled. When D_M is S/C and D_A is O/C (i. e., the mitral valve is open and aortic valve is closed) the circuit in Fig. 79.3 represents the phase when the left ventricle is filling. When D_M is O/C and D_A is S/C (i. e., the mitral valve is closed and the aortic valve is open) the circuit represents the phase when the ventricle is ejecting. The circuit in which both diodes are O/C (i. e., both the mitral and aortic valves are closed) will occur twice and represents the phase when the ventricle is undergoing isovolumic contraction and relaxation. Finally, the circuit in which both diodes are S/C (i. e., both the mitral and aortic valves are open) is clearly not feasible. The three different phases of operation of the left ventricle over four consecutive different time intervals within the cardiac cycle are summarized in Table 79.2. Every phase within the cardiac cycle is therefore modeled by a different equivalent circuit and hence a different set of linear time-varying differential equations. However, by appropriately modeling the diodes as nonlinear elements, it is possible to write only one set of differential equations, which describes the behavior of the entire model for all three phases. Selecting the state variables as the hemodynamic variables listed in Table 79.3 (and shown on the circuit in Fig. 79.2), and using basic circuit analysis methods [79.23] such as Kirchhoff's voltage and current laws (KVL and KCL), we can derive the state equations for the cardiovascular circuit model shown in Fig. 79.2 as

$$\begin{pmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \\ \frac{dx_3}{dt} \\ \frac{dx_4}{dt} \\ \frac{dx_5}{dt} \end{pmatrix} = \begin{pmatrix} \frac{-\dot{C}(t)}{C(t)} & 0 & 0 & 0 & 0 \\ 0 & \frac{-1}{R_S C_R} & \frac{1}{R_S C_R} & 0 & 0 \\ 0 & \frac{1}{R_S C_S} & \frac{-1}{R_S C_S} & 0 & \frac{1}{C_S} \\ 0 & 0 & 0 & 0 & \frac{-1}{C_A} \\ 0 & 0 & \frac{-1}{L_S} & \frac{1}{L_S} & \frac{-R_C}{L_S} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} + \begin{pmatrix} \frac{1}{C(t)} & \frac{-1}{C(t)} \\ \frac{-1}{C_R} & 0 \\ 0 & 0 \\ 0 & \frac{1}{C_A} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{R_M} r(x_2 - x_1) \\ \frac{1}{R_A} r(x_1 - x_4) \end{pmatrix} . \tag{79.4}$$

In (79.4) $\dot{C}(t) = dC(t)/dt$ and $r(x)$ represents the ramp function

$$r(\xi) = \begin{cases} \xi & \text{if} \quad \xi \geq 0 , \\ 0 & \text{if} \quad \xi < 0 . \end{cases} \tag{79.5}$$

We note that the model described above is an autonomous switched linear time-varying system over four different phases within the cardiac cycle [79.15, 16]. Its solution is oscillatory due to the cyclic nature of the terms $\dot{C}(t)$ and $1/C(t)$ in the matrices in (79.4). A plot of the expression $-\dot{C}(t)/C(t)$ over one cardiac cycle for a healthy heart is shown in Fig. 79.5. The term $1/C(t)$ represents the elastance function $E_H(t)$ whose plot is shown in Fig. 79.4.

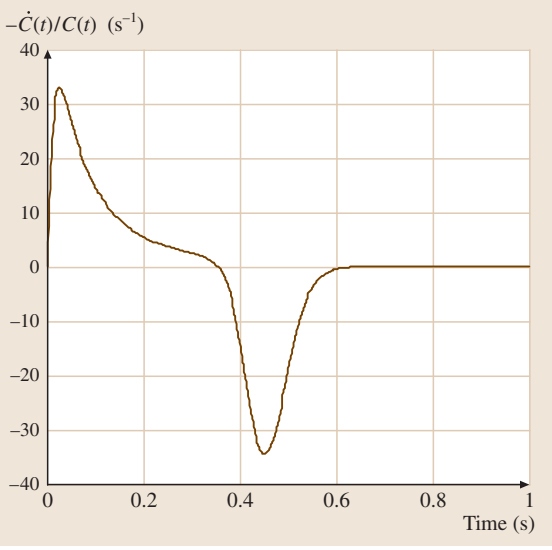


Fig. 79.5 Plot of $\frac{-\dot{C}(t)}{C(t)}$ for a healthy heart over one cardiac cycle

79.2 Cardiovascular Model Validation

The ability of the fifth-order model described in the previous section to emulate the hemodynamics of the left ventricle is demonstrated by simulating the model for both nominal steady-state conditions, and in response to perturbations of preload and afterload. Figure 79.6 shows the simulation waveforms of the hemodynamics for an adult with heart rate of 75 bpm. In this particular case, systolic and diastolic pressure were 117 and 77 mmHg, mean aortic pressure (MAP) was 99 mmHg, cardiac output (CO) (i. e., the blood pumped by the ventricle in 1 min) was 5.21 l/min and stroke volume (SV) (i. e., the volume of blood pumped by the ventricle in one cardiac cycle) was 69.5 ml/beat. These numbers and waveforms are all consistent with hemodynamic data in normal subjects described in [79.24].

A second method to validate the model and verify that it can reproduce the human left ventricle behavior is to vary the preload and afterload conditions, while keeping the left ventricle parameters (E_{max}, E_{min}, and V_0) constant. If the model behaves as expected, we should have an approximately linear relationship between end-systolic pressure and left ventricle volume (known as end-systolic pressure–volume relationship, or ESPVR), in spite of changes in preload and afterload. A total of four preload and four afterload conditions were simulated using $E_{max} = 2$ mmHg/ml, $E_{min} = 0.05$ mmHg/ml, and $V_0 = 10$ ml. The resulting pressure and volume of the ventricle are typically plotted in the form of pressure–volume (PV-)loops. The PV-loops in Fig. 79.7a, represent the result of changing afterload conditions by selecting different values of systemic vascular resistance (R_S in Fig. 79.3), while keeping end-diastolic volume (EDV) constant.

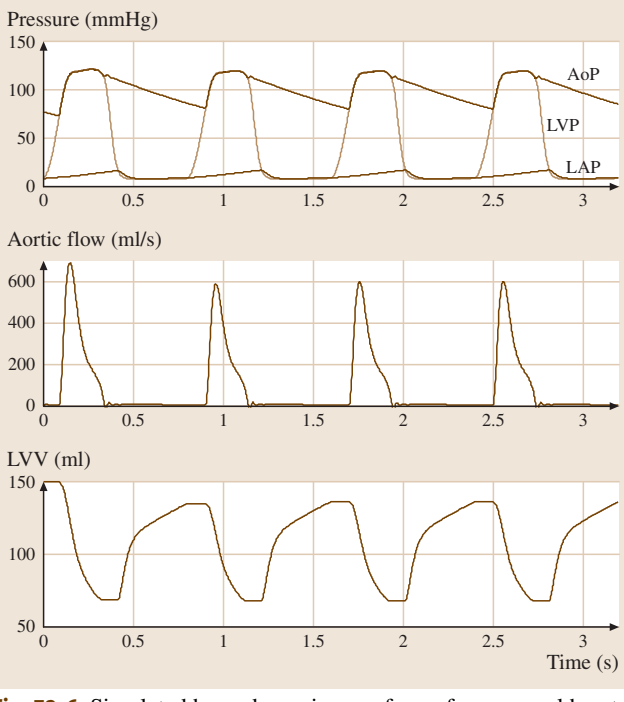


Fig. 79.6 Simulated hemodynamic waveforms for a normal heart

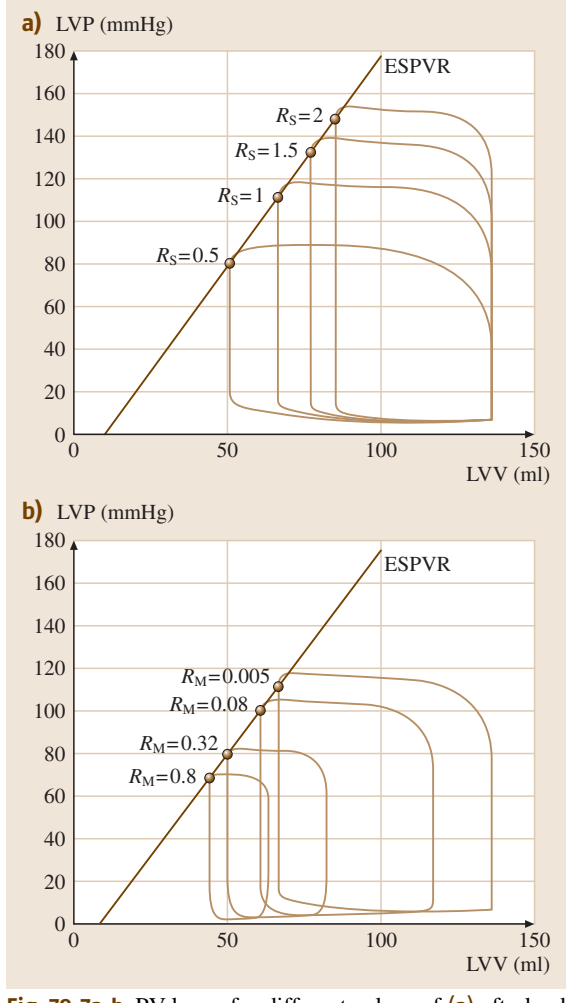


Fig. 79.7a,b PV-loops for different values of **(a)** afterload resistance, and **(b)** preload volume

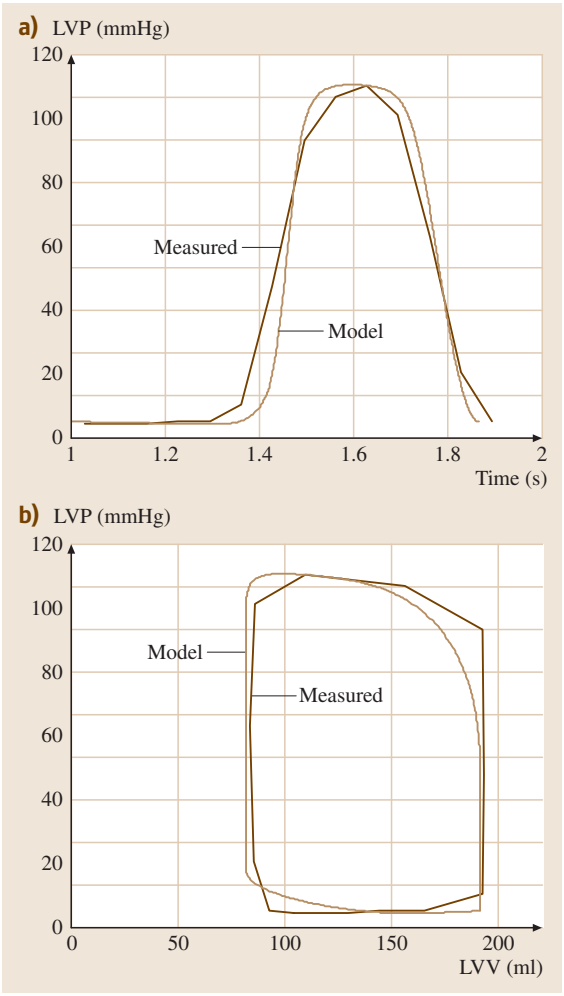


Fig. 79.8a,b Measured and simulated **(a)** left ventricular pressure and **(b)** PV-loops ▶

The PV-loops in Fig. 79.7b represent the result of altering preload conditions by changing the Mitral valve resistance R_M. The linear relationship between pressure and volume is evident in both Fig. 79.7a,b. The slope of the ESPVR and the horizontal intercept for the afterload data in Fig. 79.7a are determined as 1.98 mmHg/ml and 10.91 ml, respectively. For the preload data of Fig. 79.7b these values are determined as 1.92 mmHg/ml and 8.84 ml respectively. These results are consistent with the values used for E_{max} and V_0 used in the simulation and clearly show that the cardiovascular model can indeed mimic the behavior of the left ventricle.

Finally, a third method to validate the model is to compare the hemodynamic waveforms obtained from the model to those of a human patient. Figure 79.8 shows left ventricular pressure data and the corresponding PV-loop obtained from our model using values $E_{max} = 1.5$ mmHg/ml and $V_0 = 12$ ml estimated from real data of a patient suffering from cardiomyopathy. The measured data from the patient are also superimposed on the same plots in Fig. 79.8. Clearly the data from the model exhibit a close fit to the real measured patient data. As a test for how close the two data sets match, in the case of LVP, the error was calculated to be 4.4% and in the case of the PV-loop, the area within the loop, known as the stroke work (SW), was determined to be 10 492 mmHg · ml for the PV-loop of the model and 10 690 mmHg · ml for the patient PV-loop, which represents a difference of only 1.85%.

79.3 LVAD Pump Model

The LVAD considered in this chapter is a rotary blood pump connected using two cannulae as a bridge between the left ventricle and the aorta as illustrated in the schematic in Fig. 79.9. The pump (HeartMate II) and its cannulae are characterized by the relationships [79.25]

$$H_p = R_p Q + L_p \frac{dQ}{dt} + \beta\omega^2 , \quad (79.6)$$

$$H_i = R_i Q + L_i \frac{dQ}{dt} , \quad (79.7)$$

$$H_o = R_o Q + L_o \frac{dQ}{dt} , \quad (79.8)$$

where H_p, H_i, and H_o are the pressure differences across the pump and the inlet and outlet cannulae, respectively, Q is the flow rate of blood through the pump and cannulae, and ω is the pump rotational speed. The parameters R_p, R_i, and R_o represent the resistances and the parameters L_p, L_i, and L_o represent the inertances of the pump and cannulae, respectively. The parameter β is a pump-dependent constant. We note that R_i repre-

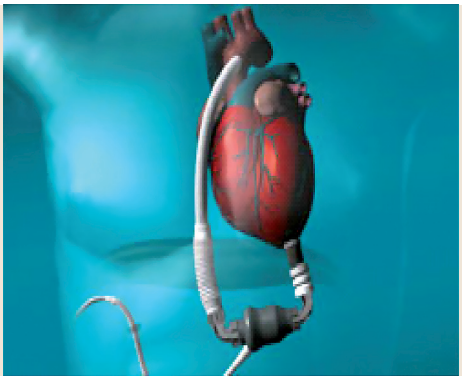


Fig. 79.9 Schematic of a rotary LVAD connected between the left ventricle and the aorta

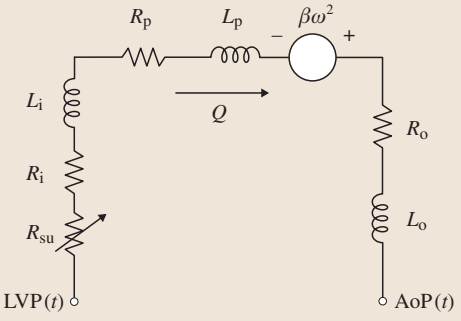


Fig. 79.10 LVAD and cannulae equivalent circuit

sents the resistance of the inlet cannula when no suction is present. When suction occurs, an additional resistance R_{su} in the form [79.5, 26]

$$R_{su} = \begin{cases} 0 & \text{if} \quad \text{LVP}(t) > \bar{x}_1 \\ \alpha(\text{LVP}(t) - \bar{x}_1) & \text{if} \quad \text{LVP}(t) \le \bar{x}_1 \end{cases} \tag{79.9}$$

is added to the expression in (79.7) to represent this phenomenon. Clearly, R_{su} is a nonlinear time-varying

Table 79.4 Model parameters for the LVAD

Parameters	Value	Physiological meaning
Cannulae resistances (mmHg s/ml)		
R_{su}	See (79.9)	Suction resistance with parameters $\alpha = -3.5$ s/ml and $\bar{x}_1 = 1$ mmHg
R_i	0.0677	Inlet cannula resistance
R_p	0.17070	Outlet cannula resistance
R_o	0.0677	Pump resistance
Cannulae inertances (mmHg s^2/ml)		
L_i	0.0127	Inlet cannula inertance
L_p	0.02177	Pump inertance
L_o	0.0127	Outlet cannula inertance
Pump parameter (mmHg/(rpm)^2)		
β	9.9025×10^{-7}	

resistance whose value is zero when the pump is operating normally and is activated when the left ventricular pressure LVP(t) (or $x_1(t)$) becomes less than a predetermined small threshold $\bar{x}_1$, a condition that represents suction. The value of R_{su} when suction occurs increases linearly as a function of the difference between LVP(t) and the threshold $\bar{x}_1$. The parameter α is a cannula-dependent scaling factor. An equivalent circuit model for the LVAD and inlet and outlet cannulae is shown in Fig. 79.10. Values of all parameters mentioned above for the specific LVAD used in our model are given in Table 79.4. The state equation governing the behavior of the LVAD can now be easily derived as

$$\text{LVP}(t) - \text{AoP}(t) = R^* Q + L^* \frac{\mathrm{d}Q}{\mathrm{d}t} + \beta\omega^2 \,, \tag{79.10}$$

where

$$R^* = R_{su} + R_i + R_p + R_o \tag{79.11}$$

and

$$L^* = L_i + L_p + L_o \,. \tag{79.12}$$

79.4 Combined Cardiovascular and LVAD Model

Figure 79.11 shows the equivalent electric circuit of the combined cardiovascular and LVAD model. The addition of the LVAD to the model represented in (79.4) adds one state variable $x_6(t) = Q$, which represents the blood flow through the pump and eight passive parameters R_{su}, R_i, R_p, R_o, L_i, L_p, L_o, and β. Note that the combined cardiovascular and LVAD model is now a forced system, where the primary control variable is the pump speed ω. As was done for the cardiovascular model, the state equations for this combined sixth-order

model can be derived from basic circuit analysis techniques and written in the form

$$\begin{pmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \\ \frac{dx_3(t)}{dt} \\ \frac{dx_4(t)}{dt} \\ \frac{dx_5(t)}{dt} \\ \frac{dx_6(t)}{dt} \end{pmatrix} = \begin{pmatrix} \frac{-\dot{C}(t)}{C(t)} & 0 & 0 & 0 & 0 & \frac{-1}{C(t)} \\ 0 & \frac{-1}{R_S C_R} & \frac{1}{R_S C_R} & 0 & 0 & 0 \\ 0 & \frac{1}{R_S C_S} & \frac{-1}{R_S C_S} & 0 & \frac{1}{C_S} & 0 \\ 0 & 0 & 0 & 0 & \frac{-1}{C_A} & \frac{1}{C_A} \\ 0 & 0 & \frac{-1}{L_S} & \frac{1}{L_S} & \frac{-R_C}{C_S} & 0 \\ \frac{1}{L^*} & 0 & 0 & \frac{-1}{L^*} & 0 & \frac{-R^*}{L^*} \end{pmatrix} \times \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \\ x_5(t) \\ x_6(t) \end{pmatrix} + \begin{pmatrix} \frac{1}{C(t)} & \frac{-1}{C(t)} \\ \frac{-1}{C_R} & 0 \\ 0 & 0 \\ 0 & \frac{1}{C_A} \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \times \begin{pmatrix} \frac{1}{R_M} r(x_2 - x_1) \\ \frac{1}{R_A} r(x_1 - x_4) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{\beta}{L^*} \end{pmatrix} u(t) \,. \tag{79.13}$$

The control variable in (79.13) is $u(t) = \omega^2(t)$, where $\omega(t)$ is the rotational speed of the pump. Using control terminology, and after some simple algebraic manipulations, this model can be expressed as a switched time-varying linear system in the standard state-space form [79.15, 16]

$$\frac{d\boldsymbol{x}}{dt} = \mathbf{A}_k(t)\boldsymbol{x} + \boldsymbol{b}u \,, \quad k = \mathrm{I}, \mathrm{F}, \mathrm{E} \,, \tag{79.14}$$

where $\boldsymbol{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^\top$ is the state vector, $\boldsymbol{b} = (0, 0, 0, 0, 0, \beta/L^*)^\top$, and $(\,)^\top$ denotes transpose. The matrix $\mathbf{A}_k(t)$ in (79.14) switches over each of the four different phases within the cardiac cycle (Table 79.2). The description of this matrix over each of these regions is given as follows.

1. *Isovolumic relaxation and contraction phases* ($k = \mathrm{I}$): Within these two intervals we have $x_2 - x_1 < 0$ and $x_1 - x_4 < 0$. The matrix $\mathbf{A}_k(t)$ will take the form

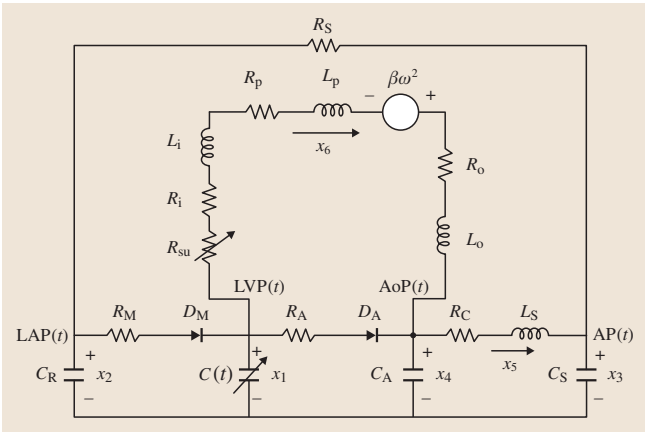


Fig. 79.11 Combined cardiovascular and LVAD model

$$\mathbf{A}_\mathrm{I}(t) = \begin{pmatrix} \frac{-\dot{C}(t)}{C(t)} & 0 & 0 & 0 & 0 & \frac{-1}{C(t)} \\ 0 & \frac{-1}{R_S C_R} & \frac{1}{R_S C_R} & 0 & 0 & 0 \\ 0 & \frac{1}{R_S C_S} & \frac{-1}{R_S C_S} & 0 & \frac{1}{C_S} & 0 \\ 0 & 0 & 0 & 0 & \frac{-1}{C_A} & \frac{1}{C_A} \\ 0 & 0 & \frac{-1}{L_S} & \frac{1}{L_S} & \frac{-R_C}{L_S} & 0 \\ \frac{1}{L^*} & 0 & 0 & \frac{-1}{L^*} & 0 & \frac{-R^*}{L^*} \end{pmatrix} . \tag{79.15}$$

2. *Filling phase* ($k = \mathrm{F}$): Within this interval we have $x_2 - x_1 \geq 0$ and $x_1 - x_4 < 0$. The matrix $\mathbf{A}_k(t)$ will take the form

$$\mathbf{A}_\mathrm{F}(t) = \begin{pmatrix} \frac{-\dot{C}(t)}{C(t)} - \frac{1}{R_M C(t)} & \frac{1}{R_M C(t)} & 0 & 0 & 0 & \frac{-1}{C(t)} \\ \frac{1}{R_M C_R} & \frac{-1}{C_R}\left(\frac{1}{R_M} + \frac{1}{R_S}\right) & \frac{1}{R_S C_R} & 0 & 0 & 0 \\ 0 & \frac{1}{R_S C_S} & \frac{-1}{R_S C_S} & 0 & \frac{1}{C_S} & 0 \\ 0 & 0 & 0 & 0 & \frac{-1}{C_A} & \frac{1}{C_A} \\ 0 & 0 & \frac{-1}{L_S} & \frac{1}{L_S} & \frac{-R_C}{L_S} & 0 \\ \frac{1}{L^*} & 0 & 0 & \frac{-1}{L^*} & 0 & \frac{-R^*}{L^*} \end{pmatrix} . \tag{79.16}$$

3. *Ejection phase* ($k = \mathrm{E}$): Within this interval we have $x_2 - x_1 < 0$ and $x_1 - x_4 \geq 0$. The matrix $\mathbf{A}_k(t)$ will

take the form

$$\mathbf{A}_{\mathrm{E}}(t) = \begin{pmatrix} \frac{-\dot{C}(t)}{C(t)} - \frac{1}{R_{\mathrm{A}}C(t)} & 0 & 0 & \frac{1}{R_{\mathrm{A}}C(t)} & 0 & \frac{-1}{C(t)} \\ 0 & \frac{-1}{R_{\mathrm{S}}C_{\mathrm{R}}} & \frac{1}{R_{\mathrm{S}}C_{\mathrm{R}}} & 0 & 0 & 0 \\ 0 & \frac{1}{R_{\mathrm{S}}C_{\mathrm{S}}} & \frac{-1}{R_{\mathrm{S}}C_{\mathrm{S}}} & 0 & \frac{1}{C_{\mathrm{S}}} & 0 \\ \frac{1}{R_{\mathrm{A}}C_{\mathrm{A}}} & 0 & 0 & \frac{-1}{R_{\mathrm{A}}C_{\mathrm{A}}} & \frac{-1}{C_{\mathrm{A}}} & \frac{1}{C_{\mathrm{A}}} \\ 0 & 0 & \frac{-1}{L_{\mathrm{S}}} & \frac{1}{L_{\mathrm{S}}} & \frac{-R_{\mathrm{C}}}{L_{\mathrm{S}}} & 0 \\ \frac{1}{L^*} & 0 & 0 & \frac{-1}{L^*} & 0 & \frac{-R^*}{L^*} \end{pmatrix} . \tag{79.17}$$

Switched linear systems have been studied extensively in recent years, especially with regards to stability and stabilization [79.27, 28]. The development of an optimal feedback controller for the above system will present a major control challenge, especially in guaranteeing that the resulting closed-loop autonomous system will also be stable.

79.5 Challenges in the Development of a Feedback Controller and Suction Detection Algorithm

In assisting the heart, the amount of blood flow required of an LVAD depends on the patient's level of activity, emotional state, posture, and other physiological demands. The response of the heart itself due to changes in demand is governed by its contractility, and the preload and afterload. Contractility of the heart is related to the intrinsic ability of the heart muscle to contract. As mentioned earlier, preload is closely related to the cardiac output (CO) produced by the heart for a given level of contractility and afterload can be described as the pressure that the left ventricle has to generate in order to eject blood. Everything else held equal, if afterload increases, cardiac output will decrease. In a healthy heart, these internal control mechanisms provide the cardiovascular system with a significant capability to respond to changes in demand for blood flow in the body.

When a rotary LVAD is implanted in a patient with congestive heart failure, the only available mechanism to control it is the rotational speed ω of the pump. Currently, patients with rotary LVADs are typically kept in a critical care setting and are continuously supervised by human operators. When the patient is in a stable condition, the operator adjusts the pump speed manually so as to achieve the desired levels of the hemodynamic variables and ensure the well being of the patient. However, for a patient whose level of activity is continuously changing, this manual open-loop control becomes impractical. Its limitations will become even more apparent as patients are rehabilitated and seek to adopt a normal active lifestyle. The inability of open-loop manual control to respond automatically to changes in demand can dramatically impact the quality of life of LVAD patients [79.3, 4]. For successful long-term implantation of these devices, the need for a human to monitor the device operation must therefore be eliminated. In order to achieve a desired level of automatic operation, a feedback controller which is capable of continuously monitoring the patient's status to recognize a change in the level of activity and automatically adjust the pump speed must be incorporated in the device. As mentioned earlier, such a controller must also meet the other objective of ensuring that suction does not occur by unnecessary excessive pumping. The achievement of these two objectives has been a major

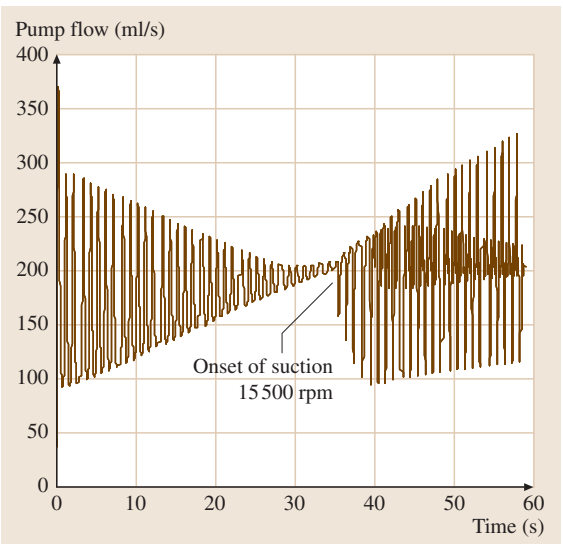


Fig. 79.12 Pump flow signal from human cardiovascular/LVAD model as a function of linearly increasing pump speed

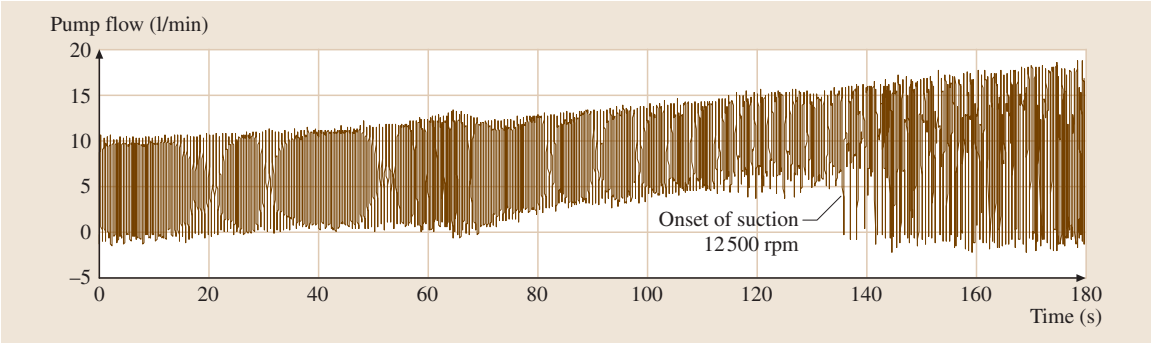


Fig. 79.13 Pump flow signal from in vivo data as a function of linearly increasing pump speed

challenge for LVAD developers for over 15 years and is recognized as one of the most serious limitations of this technology at the present time.

The design of a stable feedback controller based on the model developed earlier in this chapter is very much related to our ability to continuously measure, or estimate, in real time the patient's hemodynamic variables (x_1 through x_5 in the model). The patient's continuously varying levels of physical and emotional activities are exhibited by possibly wide variations in the systemic vascular resistance (R_s in the model), which in turn affects all six state variables in the system. Unfortunately, at present, current technology for implantable sensors that allow for continuous measurements of the patient's hemodynamic variables does not exist and will probably need many years before it can be developed. The pump flow variable x_6, on the other hand, appears to be the only variable that can be measured in real time. This may be done, for example, by using standard ultrasonic flow transducers that can be clamped onto one of the pump cannulae. With this single variable, the problem essentially reduces to a feedback controller design based on incomplete state measurements. Given that the system is nonlinear and time varying, this remains a challenge from both the theoretical and practical considerations. An attempt to develop a very simple feedback controller based on the slope of the lower envelope of the pump flow signal has recently been made [79.29] but its practical usefulness in an in vivo experiment has not yet been tested. The approach of estimating the hemodynamic variables (x_1 through x_5) so as to implement a full state feedback controller has also been considered but with limited success [79.30–34]. This is largely due to the fact that this approach involves first estimating the patient's systemic vascular parameters. Most methods developed for estimating these parameters suffer from the problem that the estimates

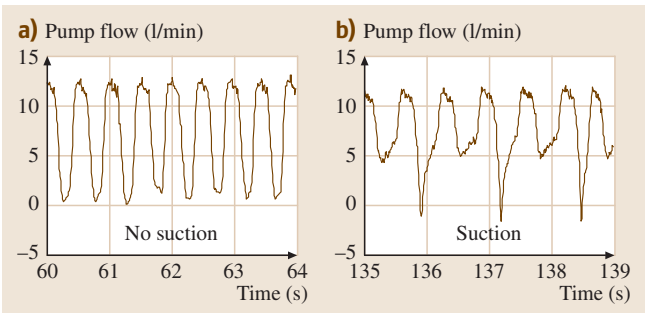


Fig. 79.14a,b In vivo pump flow signal **(a)** when pump is operating normally, and **(b)** when pump is in suction

obtained are discrete and valid only over parts of the cardiac cycle. A method for simultaneously estimating the parameters and the hemodynamic variables has been developed in [79.21, 22] based on an extended Kalman filter approach; however, the robustness of the method with respect to parameter initialization remains questionable and its usefulness is limited due to uncertainties related to convergence of the algorithm to the correct estimates. In summary, the design of a feedback controller for the LVAD remains a very challenging open problem.

Since the pump flow signal appears to be the only signal that is directly measurable, it is important to examine how this signal is affected by the pump speed as it changes over its allowable safe range. Figure 79.12 shows a plot of this signal (state variable x_6) when our model is exited with a pump speed starting at 12 000 rpm and increasing linearly according to $\omega = 12\,000 + 100t$, reaching a speed of 18 000 rpm after 60 s. It is clear from this figure that the lower envelope of this signal seems to track the increasing pump speed up to a point when a breakdown occurs and then exhibits a sudden drop when the speed is increased

beyond the breakpoint. In Fig. 79.12, this breakdown occurs at $t = 35$ s, which corresponds to a speed of about 15 500 rpm and is indicative of the onset of suction in the model. The same phenomenon has been observed in an in vivo animal study (authorized according to WorldHeart, Inc. IRB DO 01-06002) in which the WorldHeart (World Heart Inc.: Formerly MedQuest, Inc., Salt Lake City, UT) rotary LVAD was used. In this study the pump speed was increased linearly from 8000 rpm according to $\omega = 8000 + \frac{100}{3}t$ and reached a speed of 14 000 rpm after 180 s. The corresponding measured pump flow signal is shown in Fig. 79.13. Again, in this case, a similar breakdown in the lower envelope of the signal is observed at $t = 136$ s, which corresponds to a speed 12 500 rpm, indicating the onset of suction in the animal's left ventricle. It is interesting to note that, in both the model simulation and the in vivo experiment, the onset of suction appears to be characterized by a sudden large drop in the lower envelope of the pump flow signal. Furthermore, when the pump is in suction for some time, a noticeable change in the characteristics of the signature of the pump flow signal is observed. This is clearly seen in Fig. 79.14 in which the details of the in vivo data of Fig. 79.13 are shown over two 5 s intervals of time from 60 to 64 s before the occurrence of suction and from 135 to 139 s during suction. The change in the frequency characteristics of the signal from a narrow-band signal to a relatively broad-band signal is very noticeable. The breakpoint in the lower envelope of the pump flow signal and the changes in its frequency characteristics before and after suction represent opportunities that can be exploited in the development of suction detection algorithms. We should note that suction detection has been a long-standing problem that has been studied ever since the rotary LVAD was introduced. Many algorithms have been proposed based on numerous other criteria and indices [79.35–39]. A successful attempt to develop a suction detection algorithm based on the frequency characteristics of the pump flow signal has recently been reported [79.40, 41].

79.6 Conclusion

In this chapter, a sixth-order state-space model of a left ventricular assist device connected to a cardiovascular system is presented. The model is obtained by combining a fifth-order model of the hemodynamics of the cardiovascular system with a first-order model of the blood flow in a rotary pump along with its inlet and outlet cannulae. The phenomenon of suction is accounted for by including a nonlinear resistance in the model of the inlet cannula. The resulting combined model is nonlinear and time-varying with a different mathematical description over the four different time intervals which correspond to the four phases of the ventricular functions within one cardiac cycle. The only control variable in the model is the rotational speed of the pump. Given that current implantable sensor technology does not exist for measuring the patient's hemodynamic variables, the challenges in using this model to design a feedback speed controller for the LVAD are discussed. The characteristics of the pump flow signal – which is the only directly measurable variable – when the pump is operating normally with no suction and when it is operating in suction are also described based on data obtained from the model as well as from an in vivo animal experiment. Possible approaches for exploiting these characteristics in the development of suction detection algorithms are also discussed.

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